Design for Discovery

The most apparent takeaway of this project's development has been the way in which stairs and vertical circulation interact with bay sizes and fixed proportions. Going forward, that has to become a foundational part of design concepts, because it does not just work itself out efficiently.

Design for Well-Being

Every space in the project is tied back to a centralized, glass, vertical circulation core that looks out towards the Catalina Mountains. Every space has either one point of direct access or multiple points of secondary access to daylight, but is sheltered from direct sunlight.

Design for Equitable Communities

Each space in the Forum is visually and atmospherically connected to multiple others, promoting interaction and connection between users of all abilities and objectives.
Each rooftop terrace is accessible via the elevator which occupies the same column as the stairs. The cafe, event space, water labs, resource center, and library are all accessible from the first floor.

Sonoran Waters Forum

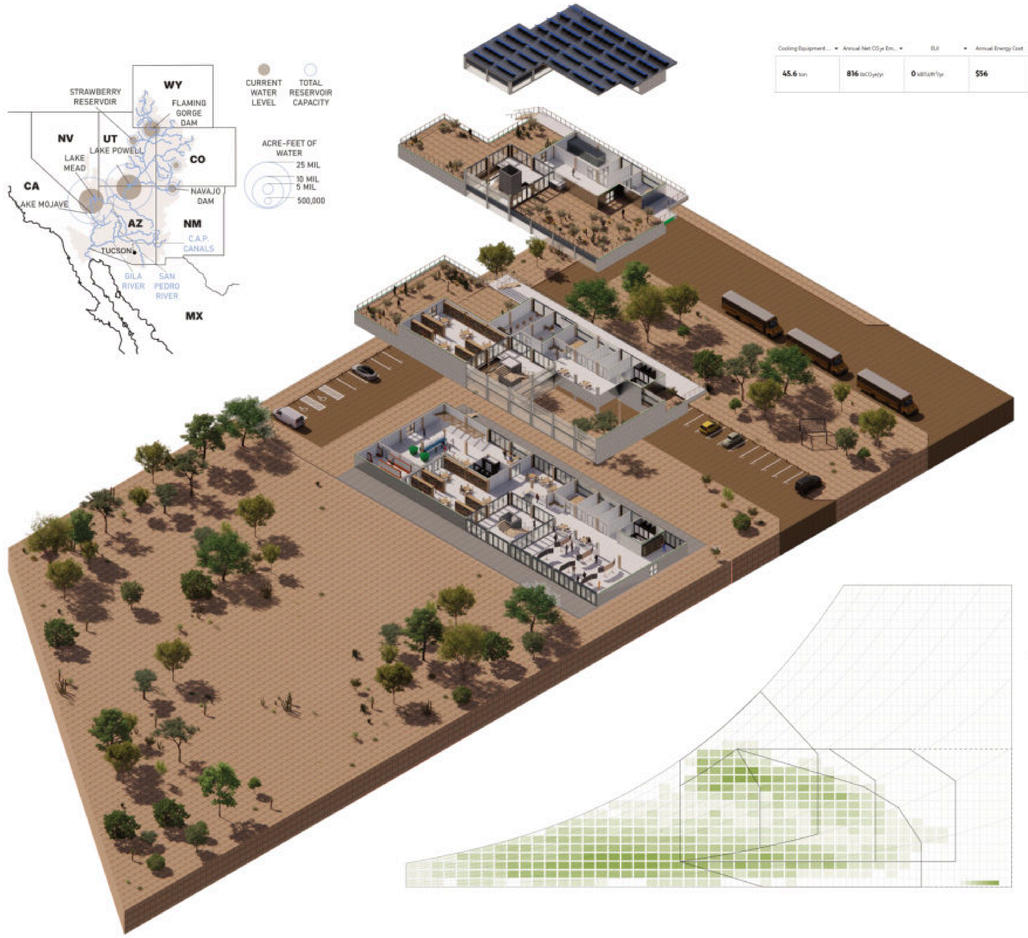
The Sonoran Waters Forum is a three story research, education, and event space, aimed at developing urban farming and exemplifying care for water scarcity and energy sustainability. In conjunction with Tucson Village Farm, the Forum serves to educate the next generation on techniques and technology of food and water as well as local plants and animals.

The goal of the client was to make a visible attraction to the site, to provide a multipurpose space for hosting events and engaging the public, and to create facilities wherein meaningful research can be conducted on natural resources, water management, and hydroponic agriculture. The goal of the building was to make a terraced set of roof gardens providing a means of gradual entry or egress, visually connect all spaces for the sake of the users, and to make a range of intimate and lively spaces.

The Sonoran Waters Forum aims to connect the immeasurable and the measurable by providing opportunities for discovery and spontaneous interaction, within the bounds of sensible arrangement. Humid spaces like the farming and water labs are sealed off from the library, but the cafe is connected directly to the event space, classrooms, and primary entrance.

Design for Water

To combat the shrinking water table, and to care for the vegetation on site, groundwater is harvested and used to offset graywater use as well as to irrigate the plant life.
Especially for the plants on the green-roof, it is invaluable to effectively steward the natural commodity that is water in the desert.



Cooling Equipment...	Annual Net CO_2e Em...	EUI	Annual Energy Cost
45.6 ton	816 $tonCO_2e/yr$	0 $kBtu/ft^2/yr$	$56

Design for Energy

The Sonoran Waters Forum is a net-zero building, producing enough electricity with its photovoltaic system to cover the remainder of all its energy consumption that is not already mitigated by efficient design, passive chilled beams, and regulated operational hours. Once built, the building is practically self-sustaining.

Design for Economy

All materials are based on a sixteen foot grid, ensuring that standard nominal lumber can be used for most if not all of its construction.
Beyond practical construction cost considerations, the building is also designed to be extremely efficient in operational costs as well. The estimated annual cost of operating the building is only fifty-six dollars.

Design for Integration

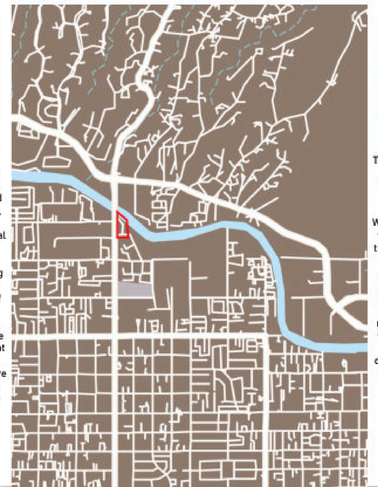
To address Tucson's massive swing in temperatures, no windows are present on the East or West facades mitigating direct summer sun.
Every exterior wall has an additional layer of rigid insulation outside the framing so as to address extreme cold and extreme heat.

Design for Resources

The structural members of the Forum are all designed to be Cross Laminated Timber, requiring no carbon intensive steel beams and columns, nor concrete beyond the foundation. All sheer walls and floor plates are made of C.L.T. sheeting, and all beams, columns, girders, and structural elements are made of mass timber instead of conventional materials.

Design for Ecosystems

Handily within the city, at the base of the mountains, above an underground river, and attached to farmlands, this site is critical for the preservation of local plant and animal life.
It is crucial that the footprint of the building does not excessively disrupt the usability of the landscape.
Additionally, because of the connection to the river, it is important that site water be utilized well, for the care and protection of the plantlife, and not go to waste.



Design for Change

The primary event space is designed to be modular and practical for any number of occupant types.
Whether it's a function of the water lab, allowing the facility users access to the provided context, or a separate thing entirely, bringing community members into contact with the mission of the building, there is no shortage of ways in which to configure and make use of the provided areas.